AWS SAA-C03 Exam Note: CloudFront & AWS Global Accelerator

1. CloudFront Overview

- **What is it?** A global Content Delivery Network (CDN) that delivers data, videos, applications, and APIs to customers globally with low latency and high transfer speeds.
- **Key Concept:** Content is cached at **Edge Locations** (points of presence) closest to the user.
- **Benefits:**
  - Improves read performance (content is closer to users).
  - Reduces load on the Origin server.
  - **DDoS Protection:** Integrated with AWS Shield (Standard is free) and AWS WAF (Web Application Firewall).
- **Origins:**
  - **S3 Bucket:** For distributing files and caching them at the edge.
  - **Custom Origin (HTTP):** Application Load Balancer (ALB), EC2 instance, S3 Website (static hosting enabled), or any non-AWS HTTP backend.

2. CloudFront - S3 Security (OAC vs OAI)

When using S3 as an origin, you must prevent users from accessing the S3 bucket directly (which would bypass CloudFront fees, WAF, and restrictions).

- **The Solution:** Use **Origin Access Control (OAC)** (or the legacy Origin Access Identity - OAI) to restrict access.
- **How it works:**
  1. Create an OAC in CloudFront.
  2. Update the **S3 Bucket Policy** to deny public access and **only allow** the CloudFront Service Principal.

OAC vs OAI (Exam Key Topic)

Feature	Origin Access Identity (OAI)	Origin Access Control (OAC)
Status	Legacy (Deprecated)	**Recommended (Modern)**
Security Protocol	Older protocol	**SigV4** (Stronger security)
SSE-KMS Support	No (Cannot serve KMS-encrypted files)	**Yes** (Supports SSE-KMS encrypted buckets)
HTTP Methods	GET, HEAD only	All (GET, PUT, POST, DELETE, etc.)

Feature	Origin Access Identity (OAI)	Origin Access Control (OAC)
Region Support	Limited in newer regions	All AWS Regions

> **Exam Tip:** If a scenario mentions "KMS encrypted S3 objects" or "uploading via CloudFront (PUT/POST)" and asks for the security solution, the answer is **OAC**.

3. CloudFront - ALB/EC2 as an Origin

When using an EC2 instance or ALB as an origin, CloudFront connects to it using the HTTP/HTTPS protocol.

- **Security Challenge:** Users might bypass CloudFront and access the ALB/EC2 directly, which bypasses WAF rules or caching logic.
- **How to Restrict Access (Exam Favorite):**
  1. **Security Groups (Network Level):** Configure the EC2 or ALB Security Group to **only** allow Inbound traffic from CloudFront's IP ranges.
     - *Tip:* Use the **AWS Managed Prefix List** for CloudFront in the Security Group source.
  2. **Custom Headers (Application Level):**
     - Configure CloudFront to add a **custom HTTP header** (e.g., `X-Custom-Secret`) to every request it sends to the origin.
     - Configure the ALB Listener Rules (or backend application) to **only** process requests that contain this specific header and value.
     - Requests without the header (direct access) are dropped or returned as 403 Forbidden.

4. CloudFront - Geo Restriction

- **Purpose:** Restrict access to your content based on the viewer's geographic location.
- **Mechanism:** Uses a third-party Geo-IP database to determine the user's country.
- **Modes:**
  - **Whitelist:** Allow access ONLY to users in specific countries.
  - **Blacklist:** Prevent access to users in specific countries.
- **Use Case:** complying with copyright laws (e.g., streaming sports only allowed in specific territories).
- **Note:** This restricts access *at the CloudFront edge*. (For finer control like restricting specific *files*, you would use Signed URLs/Cookies, not Geo Restriction).

5. CloudFront Security (AWS WAF & Shield)

CloudFront integrates seamlessly with AWS WAF to provide Layer 7 protection at the edge.

- **AWS WAF (Web Application Firewall):**
  - **Protects Against:** SQL Injection, Cross-Site Scripting (XSS), and common web exploits.
  - **Scope:** You create a **Web ACL** and associate it with the CloudFront distribution (Global resource).
  - **WAF Geo Match vs. CloudFront Geo Restriction:**
    - **CloudFront Geo Restriction:** "All or Nothing." You block a country for the *entire* distribution.
    - **WAF Geo Match:** Granular control. You can block a country only for specific paths (e.g., `/admin`) or combine it with other rules.
- **AWS Shield:**
  - **Shield Standard:** Automatically enabled for all CloudFront customers (Free). Protects against common Layer 3/4 DDoS attacks.
  - **Shield Advanced:** Paid service ($3k/mo). Provides specialized support, cost protection, and advanced reporting for sophisticated attacks.

6. CloudFront - Price Classes

CloudFront pricing varies by edge location. To save costs, you can exclude expensive regions.

- **Price Class All (Default):**
  - Uses **all** Edge Locations globally.
  - **Pros:** Best performance worldwide.
  - **Cons:** Most expensive.
- **Price Class 200:**
  - Uses most regions but **excludes the most expensive** ones.
  - *Typically excludes:* South America, Australia.
- **Price Class 100:**
  - Uses **only the cheapest** regions.
  - *Includes:* United States, Canada, Europe.
  - **Pros:** Cheapest option.
  - **Cons:** Higher latency for users in Asia, South America, etc.

7. CloudFront - Cache Invalidation

- **The Problem:** When you update a file on the origin (e.g., `logo.png`), CloudFront continues to serve the old cached version until the TTL (Time To Live) expires.
- **The Solution (Invalidation):** You can force CloudFront to remove the object from the cache immediately.
- **How it works:** You specify the path (e.g., `/images/logo.png` or `/images/*`) to invalidate.

- **Cost:** The first 1,000 invalidation paths per month are free; you pay for each one after that.
- **Better Alternative (Versioning):** instead of invalidating, use file versioning (e.g., `logo_v1.png`, `logo_v2.png`). This forces the client to fetch the new file immediately with no cost.

8. AWS Global Accelerator - Overview

- **What is it?** A networking service that improves the availability and performance of your applications for local and global users.
- **How it works:**
  - Traffic enters the AWS network at the Edge Location closest to the user.
  - It travels over the **AWS internal global backbone network** (instead of the public internet) to reach your application endpoints.
- **Key Features:**
  - **2 Anycast Static IPs:** You get 2 static IP addresses that work globally. (Great for whitelisting).
  - **Non-HTTP Support:** Works with TCP and UDP (CloudFront is HTTP/S only).
  - **No Caching:** It does *not* cache content; it optimizes the network path.
  - **Fast Regional Failover:** If one region goes down, Global Accelerator redirects traffic to a healthy endpoint in another region in less than 1 minute (DNS-based failover can take much longer due to TTL).
- **Endpoints:** Network Load Balancers (NLB), Application Load Balancers (ALB), EC2 Instances, or Elastic IPs.

Summary Table: CloudFront vs. Global Accelerator

Feature	Amazon CloudFront	AWS Global Accelerator
Primary Goal	Content Delivery & Caching	Network Performance & Path Optimization
Protocol	HTTP / HTTPS	TCP / UDP (and HTTP/S)
Caching	**Yes** (Edge Caching)	**No** (Pass-through)
IP Address	Dynamic (changes)	**2 Static Anycast IPs**
Best For	Static content (images/video), Dynamic Web	Gaming (UDP), IoT (MQTT), VoIP, HTTP APIs
Network	AWS Edge Network	AWS Global Backbone Network